1. Data Overview

For this assignment, I selected the sales dataset provided in the Excel workbook. I chose this dataset because it represents a realistic business situation where a company records sales transactions from different branches, cities, customer types, products, and product categories. From a business analytics perspective, this dataset is useful because it allows me to understand not only total sales revenue, but also how different customer groups and product categories contribute to business performance.

The original dataset contains 253 sales transaction records and 8 main columns. These columns are sale_id, branch, city, customer_type, product_name, product_category, quantity, and total_price.

Column	Description
sale_id	Unique ID for each sales transaction
branch	The branch where the sale was made
city	The city where the transaction occurred
customer_type	Type of customer, such as Member, Normal, or Unknown
product_name	Name of the product sold
product_category	Category of the product
quantity	Number of units sold
total_price	Total value of the sales transaction

The dataset includes both categorical and numerical variables. Categorical variables such as city, customer_type, and product_category are useful for comparing different business segments. Numerical variables such as quantity and total_price help measure sales volume and revenue performance.

In this analysis, I focused on understanding the structure of the dataset, cleaning data quality problems, calculating descriptive statistics, and developing two business insights. The main questions I wanted to answer were:

1. Which product categories contribute the most to total revenue?
2. Which customer type creates stronger sales performance?

I selected these questions because they are directly connected to business decision-making. If a company understands which categories and customer groups perform better, it can improve inventory planning, promotion strategy, and customer relationship management.

sale_id	branch	city	customer_typ	product_nam	product_categor	quantity	total_pric
1	A	New York	Member	Shampoo	Personal Care	3	17,66
2	B	Los Angeles	Normal	Notebook	Stationery	10	29,43
3	A	New York	Member	Apple	Fruits	15	19,26
4	A	Chicago	Normal	Detergent	Household	5	41,73
5	B	Los Angeles	Member	Orange Juice	Beverages	7	26,22
6	A	Chicago	Normal	Shampoo	Stationery	9	108,24
7	A	Chicago	Normal	Shampoo	Personal Care	1	11,46
8	B	Los Angeles	Normal	Shampoo	Household	9	175,55
9	A	Chicago	Member	Apple	Fruits	20	302,81
10	B	Los Angeles	Member	Shampoo	Fruits	19	374,48
11	A	Chicago	Normal	Detergent	Beverages	7	69,81
13	A	New York	Normal	Orange Juice	Household	4	14,08
12	B	Los Angeles	Member	Orange Juice	Household	12	88,47
13	A	New York	Normal	Orange Juice	Household	4	14,08
14	B	Los Angeles	Member	Apple	Fruits	5	47,13
15	B	Los Angeles	Normal	Apple	Beverages	3	62,53
16	B	Los Angeles	Normal	Notebook	Stationery	8	47,51
17	B	Los Angeles	Member	Apple	Personal Care	2	4,56
18	B	Los Angeles	Normal	Notebook	Fruits	15	212,82
19	A	New York	Member	Apple	Beverages	2	41,92
20	A	Chicago	Normal	Detergent	Personal Care	11	59,75
21	B	Los Angeles	Normal	Shampoo	Personal Care	11	51,32
22	B	Los Angeles	Normal	Orange Juice	Beverages	1	7,28
23	A	New York	Member	Shampoo	Beverages	16	33,38
24	A	Chicago	Normal	Apple	Fruits	2	40,77
25	B	Los Angeles	Member	Detergent	Beverages	18	218,22
26	B	Los Angeles	Normal	Notebook	Fruits	16	238,65
26	B	Los Angeles	Normal	Notebook	Fruits	16	238,65
27	A	New York	Member	Detergent	Household	2	20,44
28	A	New York	Member	Detergent	Personal Care	17	299,41
29	B	Los Angeles	Member	Shampoo	Stationery	17	223,74
30	A	Chicago	Member	Orange Juice	Unknown	15	198,54
31	A	Chicago	Unknown	Shampoo	Fruits	3	11,88
32	A	New York	Member	Orange Juice	Unknown	3	17,3
33	B	Los Angeles	Normal	Notebook	Stationery	17	126,24
34	B	Los Angeles	Normal	Orange Juice	Household	9	78,29
35	A	New York	Unknown	Apple	Personal Care	2	28,72
36	A	New York	Member	Detergent	Personal Care	15	210,9
37	B	Los Angeles	Member	Shampoo	Fruits	16	190,72
38	B	Los Angeles	Member	Notebook	Beverages	14	81,19
40	A	New York	Normal	Apple	Fruits	11	13,77
39	A	New York	Member	Notebook	Beverages	15	246,21
40	A	New York	Normal	Apple	Fruits	11	13,77
41	A	New York	Member	Notebook	Household	10	115,56
42	A	New York	Normal	Apple	Household	12	140,98
43	A	New York	Normal	Notebook	Fruits	2	8,92
44	A	Chicago	Unknown	Detergent	Fruits	11	43,08
45	B	Los Angeles	Member	Shampoo	Household	8	134,65
46	A	New York	Member	Shampoo	Fruits	8	136,53
47	A	New York	Normal	Shampoo	Stationery	8	10,96
48	B	Los Angeles	Normal	Detergent	Personal Care	20	213,57
49	B	Los Angeles	Normal	Detergent	Unknown	2	13,5
50	A	Chicago	Normal	Apple	Stationery	19	335,45
51	A	Chicago	Normal	Detergent	Stationery	19	346,02
52	A	Chicago	Member	Orange Juice	Fruits	5	59,71

2. Data Cleaning

Before analyzing the dataset, I cleaned and checked the data carefully. I did this because inaccurate data can lead to wrong business conclusions. For example, if duplicated transactions are kept in the dataset, the revenue will be overstated and the business may think sales performance is better than it actually is.

Missing values

I first checked missing values in the main columns. I used Excel filters and missing value checks to see whether any cells were blank. The result showed that there were no blank missing values in the eight main columns.

Missing Vaues	
sale_id	0
branch	0
city	0
customer_type	0
product_name	0
product_category	0
quantity	0
total_price	0

Although there were no blank cells, I noticed that some records were labelled as Unknown. This is important because "Unknown" is not technically blank, but it still shows that the data is incomplete in meaning.

Customer Type

For customer_type, there were 3 records marked as Unknown. I did not delete these rows because the transactions still have valid values for city, product, quantity, and total price. If I deleted them, I would lose real sales information.

Instead, I kept "Unknown" as a separate category. This is a better decision because it allows the business to keep the revenue record while still recognizing that customer classification was incomplete. In a real business, this could happen when the cashier forgets to select the customer type or when the customer information is not recorded properly.

	A	B	C	D	E	F	G	H
1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
34	31	A	Chicago		Shampoo	Fruits	3	11,88
38	35	A	New York		Apple	Personal Care	2	28,72
48	44	A	Chicago		Detergent	Fruits		43,08
255								
256								

1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
34	31	A	Chicago	Unknown	Shampoo	Fruits	3	11,88
38	35	A	New York	Unknown	Apple	Personal Care	2	28,72
48	44	A	Chicago	Unknown	Detergent	Fruits		43,08
255								

Product Category

For product_category, there were 6 records marked as Unknown. Similar to customer type, I decided not to remove these records because they still represent real sales transactions. However, I treated them separately in the analysis so that they would not be mixed with known categories such as Fruits, Beverages, Stationery, Household, and Personal Care.

This step is important because product category analysis is used for business decisions such as inventory planning and promotion strategy. If unknown categories are ignored, the total revenue may become inaccurate. If they are mixed into another category without evidence, the analysis may become misleading.

1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
33	30	A	Chicago	Member	Orange Juice		15	198,54
35	32	A	New York	Member	Orange Juice		3	17,3
53	49	B	Los Angeles	Normal	Detergent		2	13,5
72	68	A	New York	Normal	Detergent		1	7,22
91	87	A	Chicago	Member	Shampoo		2	5,86
103	99	A	New York	Normal	Detergent		17	277,4
255								
256								

1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
33	30	A	Chicago	Member	Orange Juice	Unknown	15	198,54
35	32	A	New York	Member	Orange Juice	Unknown	3	17,3
53	49	B	Los Angeles	Normal	Detergent	Unknown	2	13,5
72	68	A	New York	Normal	Detergent	Unknown	1	7,22
91	87	A	Chicago	Member	Shampoo	Unknown	2	5,86
103	99	A	New York	Normal	Detergent	Unknown	17	277,4
255								

Quantity

For quantity, I checked whether there were blank or abnormal missing values. There were no missing values in the quantity column. I also calculated the median quantity using Excel:

=MEDIAN(G:G) = 11

The median quantity is useful because if quantity had missing values, the median would be a reasonable replacement method. The median is less affected by extreme values than the mean. In this dataset, no replacement was needed, but checking the median helped me understand the central purchasing behavior of customers.

	B	C	D	E	F	G	H	I
1	branch	city	customer_type	product_name	product_category	quantity	total_price	
22	A	Chicago	Normal	Detergent	Personal Care		59.75	
48	A	Chicago	Unknown	Detergent	Fruits		43.08	
65	A	New York	Member	Detergent	Household		20.09	
255								

	B	C	D	E	F	G	H	I
1	branch	city	customer_type	product_name	product_category	quantity	total_price	
22	A	Chicago	Normal	Detergent	Personal Care	11	59.75	
48	A	Chicago	Unknown	Detergent	Fruits	11	43.08	
65	A	New York	Member	Detergent	Household	11	20.09	
255								

Duplicate Rows

After checking missing values, I checked for duplicate rows using Excel's Remove Duplicates function. I selected the full transaction table so that Excel would detect exact duplicate transaction records.

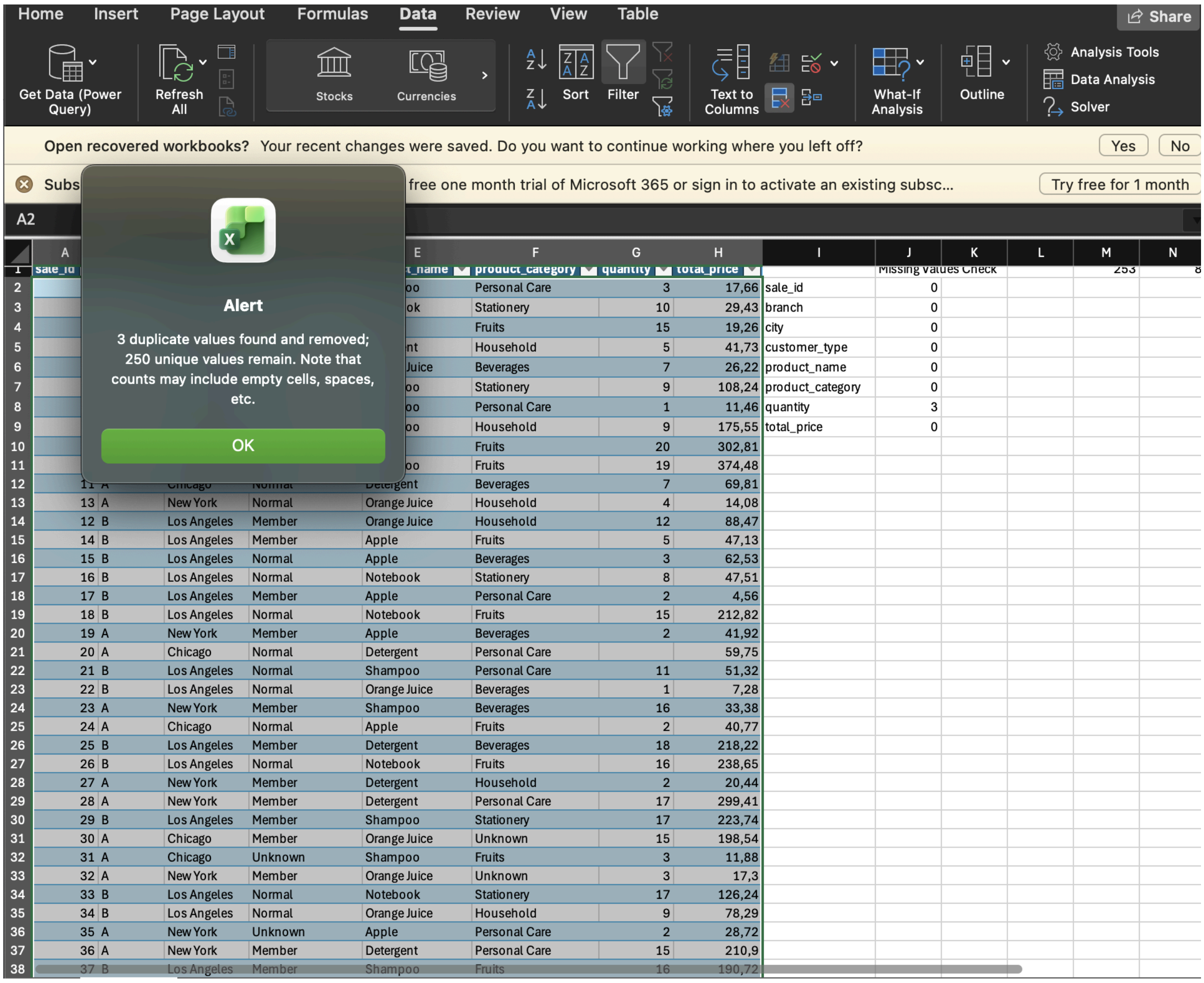
The dataset originally had 253 records. After checking duplicates, I found 3 exact duplicate rows. These duplicate transactions were related to the following sale IDs:

Duplicate sale_id	Reason
13	Exact duplicate transaction
26	Exact duplicate transaction
40	Exact duplicate transaction

After removing duplicates, the final cleaned dataset had 250 records.

Cleaning Result	Value
Original records	253
Duplicate records removed	3
Final cleaned records	250
Revenue before cleaning	31,312.78
Revenue after cleaning	31,046.28
Revenue difference	266.50

The revenue difference after removing duplicates is 266.50. This may not look very large compared with total revenue, but it still matters because duplicated data makes the business performance look better than it really is. In real business reporting, even small data errors can affect decisions about stock, staff, sales targets, and promotions.



3. Descriptive Statistics

After cleaning the dataset, I calculated descriptive statistics for the two main numerical variables: quantity and total_price.

quantity		*total_price*	
Mean	10,61264822	Mean	123,7659289
Standard Error	0,375926392	Standard Error	6,481325292
Median	11	Median	94,17
Mode	2	Mode	14,08
Standard Deviation	5,97947532	Standard Deviation	103,0917898
Sample Variance	35,7541251	Sample Variance	10627,91712
Kurtosis	-1,257327406	Kurtosis	0,033590636
Skewness	-0,072949934	Skewness	0,907731819
Range	19	Range	424,96
Minimum	1	Minimum	2,18
Maximum	20	Maximum	427,14
Sum	2685	Sum	31312,78
Count	253	Count	253

The average quantity per transaction is 10.62, while the median quantity is 11. This means the typical transaction contains around 10 to 11 units. Since the mean and median are close, quantity is quite balanced around the middle.

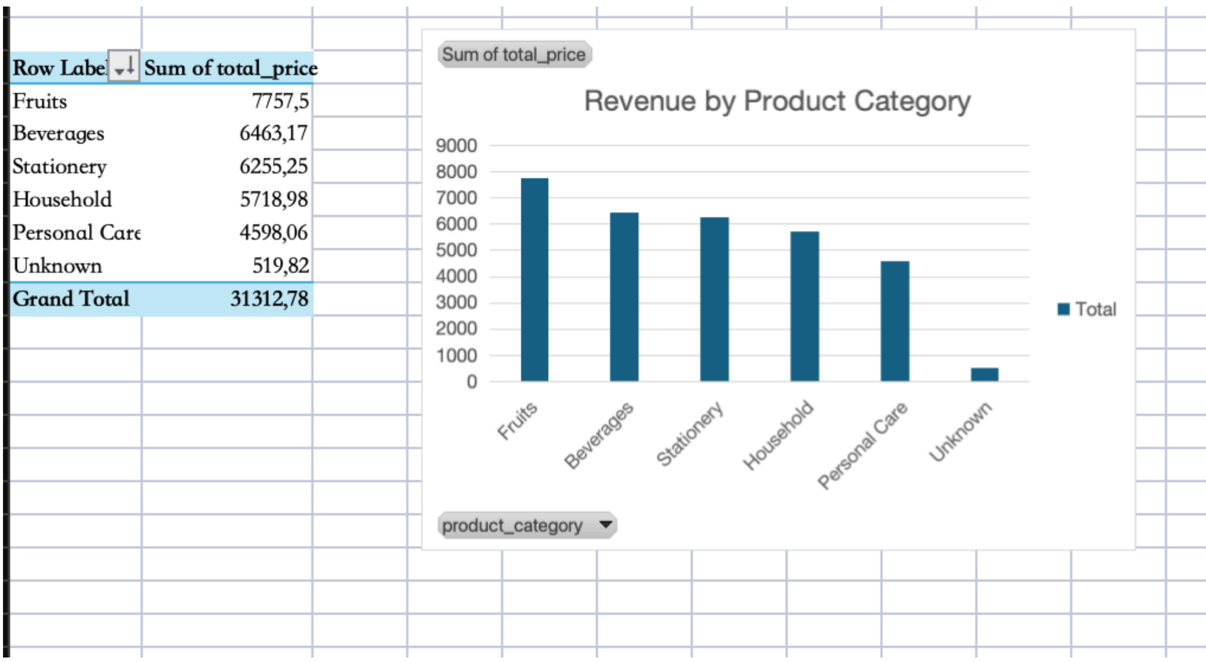
However, the total_price column shows a different pattern. The mean total price is 124.19, while the median is only 95.43. This means some high-value transactions are pulling the average upward. In other words, many transactions are around or below 95, but some much larger purchases increase the overall average.

The standard deviation of total price is 102.98, which is high compared with the mean. This shows that transaction values vary widely. From a business point of view, this means not all customers spend similarly. Some customers make small purchases, while others make much larger purchases that have a stronger impact on revenue.

4. Insights

Insight 1: Revenue by Product Category

To understand product performance, I created a PivotTable and chart showing revenue by product category.



Row Labels	Sum of total_price
Fruits	7757,5
Beverages	6463,17
Stationery	6255,25
Household	5718,98
Personal Care	4598,06
Unknown	519,82
Grand Total	31312,78

Based on the cleaned data, Fruits generated the highest total revenue, with 7,505.08, accounting for 24.17% of total revenue. This shows that Fruits is the strongest category in terms of total revenue contribution. In a real business situation, this category should be managed carefully because stock shortages in Fruits could directly reduce total sales.

However, total revenue alone does not show the full picture. Stationery had the highest average revenue per transaction, at 142.16, even though it did not generate the highest total revenue. Beverages also performed strongly, with an average revenue per transaction of 137.51. This suggests that Fruits is important for total revenue volume, while Stationery and Beverages are valuable because customers tend to spend more per transaction in these categories.

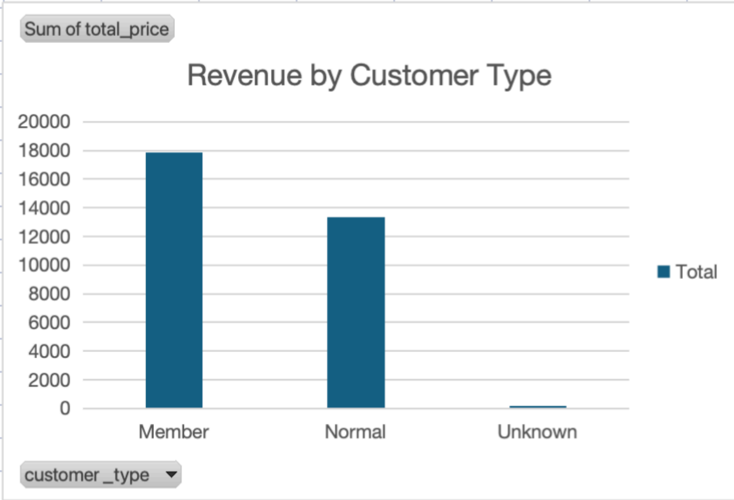
This insight helped me understand that a business should not only look at the category with the highest total sales. If the company only focuses on Fruits, it may miss opportunities to increase transaction value through Stationery and Beverages. A practical decision would be to maintain stable inventory for Fruits while using Stationery and Beverages for bundle promotions or cross-selling strategies. For example, Stationery can be promoted during school seasons, while Beverages can be paired with other frequently purchased products.

Therefore, the main business meaning of this insight is that different product categories play different roles. Fruits support total revenue, while Stationery and Beverages may help increase average basket value.

Insight 2: Revenue by Customer Type

The second analysis focuses on customer type. I created a PivotTable and chart to compare total revenue from Member, Normal, and Unknown customers.

Row Label	Sum of total_price
Member	17855,87
Normal	13373,23
Unknown	83,68
Grand Total	31312,78



The analysis shows that Member customers generated 17,855.87, which accounts for 57.51% of total revenue. Normal customers generated 13,106.73, equal to 42.22% of total revenue. This means that Member customers are the stronger customer group in this dataset.

The difference becomes clearer when looking at average revenue per transaction. Member customers spent an average of 136.30 per transaction, while Normal customers spent 112.99. This means Member customers spent about 20.64% more per transaction than Normal customers. This is an important finding because it suggests that membership is connected with stronger purchasing behavior.

From a practical business perspective, this insight supports the value of a membership or loyalty program. If members spend more per transaction, the company should not only maintain existing members but also encourage Normal customers to become members. This could be done through reward points, member-only discounts, birthday vouchers, or benefits after repeated purchases.

At the same time, the business should improve customer data recording. Although the Unknown customer type contributes only 0.27% of total revenue, it still shows that some customer information was not captured properly. If this issue continues, future customer analysis may become less accurate. Therefore, the company should make customer type selection clearer or required in the sales system.

Overall, this insight shows that customer segmentation is useful for business decision-making. Member customers are not only contributing more total revenue, but also spending more per transaction. This makes them an important group for customer retention and marketing strategy.